

Predicting Showjumping Outcomes with Self-Supervised Video Representations

CS 131 Project Milestone: Type-Conditioned Approach Quality from Unlabeled Footage

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Technical Progress

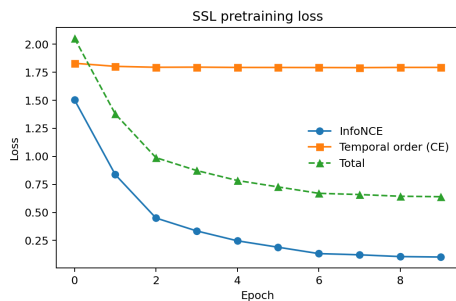
I built the milestone-stage pipeline end-to-end on Colab Pro (A100): YouTube scraping, 2-second clip segmentation, YOLOv8 horse-presence filtering, hand-annotated fence boxes, geometric takeoff-distance extraction, and a self-supervised R(2+1)D-18 video encoder with InfoNCE contrastive and 6-way temporal-order objectives. Claude Code was used to write the in-Colab annotator, to write the YouTube scraping script, and to write code documentation (comments and ReadMe). The downstream classifier and regressor heads from the proposal are not yet trained because they wait on the week-3 expansion of the labeled set. All results below use encoder embeddings, hand-drawn fence boxes, and the geometric d estimator.

Data. I scraped 6 full 1.50 m Grand Prix competitions at multiple venues (Wellington International, Tryon, Club de Campo Villa de Madrid), totaling roughly 18.5 hours of footage. A 30-second-stride window cutter produced 2,223 raw 2-second clips at 16 fps, 320×320 . A YOLOv8 horse-presence filter (keep clips with a horse visible in $\geq 40\%$ of sampled frames) reduced this to 1,121 jumping clips and moved the rest (crowd shots, course walks, slow-motion replays, interviews) into a rejected pool for inspection. From these, I hand-annotated 39 fence bounding boxes (22 oxers, 17 verticals) using an inline matplotlib + ipywidgets annotator built for Colab; each box was drawn on a frame I scrubbed to manually rather than the clip midpoint, because broadcast cameras track the horse, not the fence.

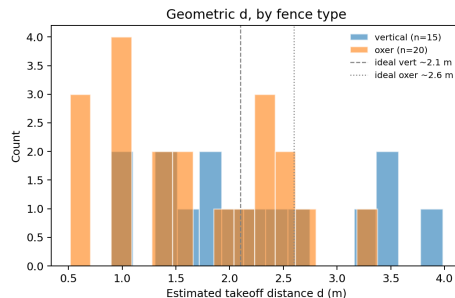
Detection and Geometry. Pretrained COCO YOLOv8 (the *horse* class) detects the horse every 2nd frame. The takeoff frame is found as the sharpest downward-to-upward reversal of the horse-box bottom edge in the second half of each clip, a robust proxy for the hoof leaving the ground. The standard 3.5 m top pole on the fence box gives meters-per-pixel as $mpp = 3.5/w_{\text{fence}}$. Takeoff distance d is then the Euclidean ground distance between the takeoff-frame horse-box bottom-center (hoof line) and the fence base, scaled by mpp. Fence type (vertical vs. oxe) is currently a width-to-height heuristic with a pole-count override from the annotator, to be replaced in week 3 by a small CNN once ≥ 50 typed labels are available.

Self-supervised pretraining. Encoder: torchvision R(2+1)D-18 trunk with a 2-layer MLP projector to 128-dim. Two pretext objectives are trained jointly: InfoNCE on two photometric- and flip-augmented views of the same clip ($\tau = 0.1$), with the rest of the batch as negatives, and 6-way classification of the permutation applied to three disjoint 4-frame sub-clips, loss weight $\lambda_{\text{order}} = 0.3$. Optimizer: AdamW (lr 10^{-3} , weight decay 10^{-4}), cosine schedule, batch 16, 10 epochs on A100.

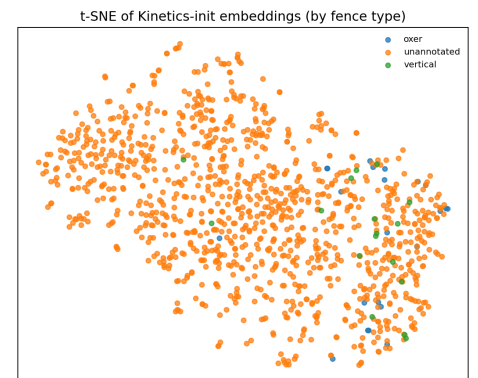
Results and Visualizations



(a) SSL pretraining losses across 10 epochs. Both InfoNCE and temporal-order CE drop monotonically; order drops below the log 6 ≈ 1.79 chance floor.



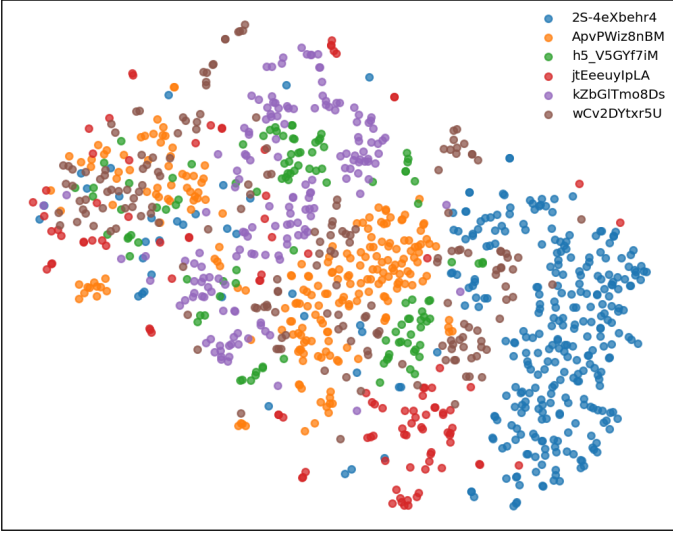
(b) Geometric d across the 39 annotated clips, split by fence type. Most values fall in the 1–3 m physically reasonable range; the > 3.5 m tail comes from oblique camera angles.



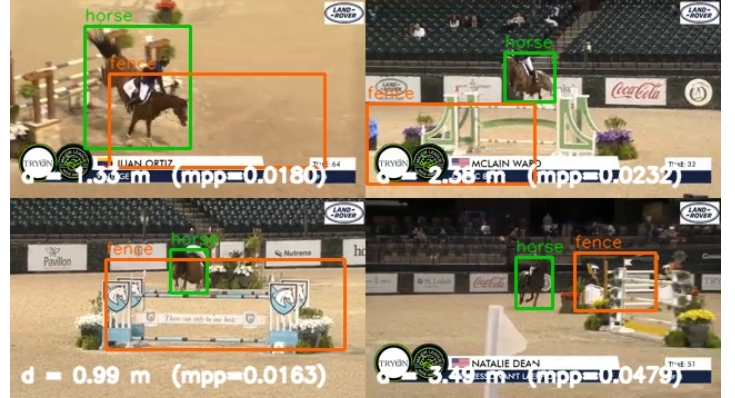
(c) t-SNE of encoder embeddings colored by hand-annotated fence type. Colored points show no clean vertical/oxer separation; most points are orange (un-annotated).

Figures. Figure 1a confirms both SSL objectives optimize cleanly, so the encoder is picking up motion direction rather than appearance. Figure 1b shows the geometric estimator produces values in the physically reasonable 1–3 m range. Figure 1c shows the embedding does *not* cleanly separate verticals from oxers, but Figure 2a explains that the

t-SNE colored by source video (venue clustering)



(a) Same t-SNE, recolored by source broadcast. Clear per-video clusters, the dominant axis of variation in the embedding space is venue (arena, lighting, footing, crowd) instead of approach quality to be addressed in the next two weeks.



(b) Four sample clips with horse box (green), hand-annotated fence box (orange), and the computed d value overlaid at the detected takeoff frame. The bottom-right panel illustrates the oblique-angle failure mode.

dominant variation is which broadcast a clip came from. This is the key empirical finding of the milestone. Figure 2b shows representative geometric successes and the main failure mode (d inflation under oblique angles).

Deviations from the Proposal

SSL collapsed from scratch. A 10-epoch run on 1,121 clips drove both losses down monotonically but yielded representational collapse (nearest-neighbor cosine similarity ≈ 0.97 across unrelated clips). I executed the proposal’s planned Kinetics-400 fallback (the pretrained R(2+1)D-18 encoder used directly) and the figures above all use these embeddings.

Venue confound. All 39 hand-annotated fences are from one video, picked for camera quality during a single annotation session. Combined with the venue-clustered embedding space, this blocks any approach-quality claim until the labeled set spans venues.

Fence detection. The proposal said YOLOv8 would label fence type from box shape and pole count; pretrained COCO YOLO has no fence class, so I hand-drew boxes for the milestone. Week 3 fine-tunes YOLO on these 39 boxes (plus new ones) to recover the original automated path.

Stretch goal deferred. Future-frame embedding prediction is not useful before the venue confound is fixed.

Updated Timeline and Plan

1. **Week 3 (Cross-venue labels & SSL fix).** Annotate ~ 110 more fence boxes spanning all 6 videos (target: 150 total). Fine-tune YOLOv8 on those boxes for automatic detection. Re-run SSL pretraining from Kinetics initialization with a venue-balanced sampler; add a domain-adversarial head to actively suppress venue identity. Train the proposal’s three baselines (from-scratch supervised, ImageNet 2D-CNN, type-only logistic regression).
2. **Week 4 (Analysis & writeup).** Sample-efficiency curves at $n \in \{25, 50, 100, 200\}$, ablations on each pretext task and on type conditioning, d -MAE reported per fence type, t-SNE colored by outcome / d / type, write final report and presentation.

Risks watched. Cross-venue annotation is the rate-limiter. Domain-adversarial training can be unstable on small datasets, so I will fall back to a simpler arena-color jitter augmentation if it diverges. Side-view camera filtering will reduce the labeled-clip pool further; the clip-stride is tunable to compensate.

Link to Codebase.: <https://github.com/bballhaus/showjumping-ssl>